

JESSICA DICHAKDJIAN

Robotics Engineer | Robot Learning & Manipulation

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PROFESSIONAL SUMMARY

Robotics & Mechatronics MSc candidate at Politecnico di Milano, currently conducting thesis research at University College London (UCL) in robot learning, manipulation, and shared autonomy. Hands-on experience spanning MuJoCo simulation, ROS-based and XR teleoperation, robot kinematics and control, embedded systems, CAD, rapid prototyping, and physical robot validation. Seeking UK robotics roles from 2027 in manipulation, robot learning, autonomy, teleoperation, control, and robotics systems integration.

TECHNICAL SKILLS

- **Programming & Robotics Software:** Python, C/C++, C#, MATLAB, ROS Noetic, ROS 2, Linux, Git
- **Robot Learning – Current Research:** PyTorch, Vision-Language-Action Models, Diffusion Policies, Gaussian Processes, Imitation Learning, Shared Autonomy
- **Simulation & Manipulation:** MuJoCo, Franka Emika Panda, MATLAB/Simscape, Forward/Inverse Kinematics, Jacobians, Dynamics, Motion Planning
- **Teleoperation & HRI:** Unity, Meta Quest / XR, MoveIt, BioIK, Human-in-the-Loop Control, Assist-as-Needed Control
- **Mechanical & Embedded Systems:** SolidWorks, CAD, 3D Printing, Arduino, Raspberry Pi, Sensors, Actuators, Rapid Prototyping, Experimental Testing

RESEARCH & ROBOTICS PROJECTS

Uncertainty-Aware Shared Autonomy for Robotic Manipulation

Aug 2026 – Present

Visiting MSc Thesis Researcher – University College London (UCL)

London, UK

- Conducting MSc thesis research on shared autonomy for Franka manipulation in MuJoCo, investigating how learned robot policies can identify uncertain states and request targeted human intervention.
- Integrating and evaluating Vision-Language-Action models within an existing UCL Franka manipulation benchmark; current work focuses on the VLA evaluation pipeline and robot-learning baseline.
- Thesis pipeline combines diffusion-policy learning, Gaussian Process-based uncertainty estimation, and XR-based corrective demonstrations for human-in-the-loop policy improvement.

GitHub: [shared-autonomy-robot-learning](#)

XR Teleoperation of a Physical Braccio Robot

MSc Academic Project

- Developed and physically validated an XR teleoperation pipeline connecting Meta Quest tracking to a Braccio manipulator through Unity, inverse kinematics, ROS, and Arduino.
- Evaluated **two control architectures**: an initial ROS/MoveIt/RViz pipeline and a lower-latency Unity/BioIK architecture after real-time delay was observed in the ROS-based implementation.
- Integrated Unity, ROS Noetic, MoveIt, BioIK, Arduino, and physical hardware into an end-to-end human-to-robot control system.

GitHub: [xr-braccio-teleoperation](#)

Industrial Robotics – Manipulation, Dynamics & Visor Automation

MSc Academic Project

- Designed and simulated a custom **4-DOF Rz–Ry–Ry–Ry** manipulator for helmet visor opening and inspection, including FK, IK, Jacobians, workspace, manipulability, dynamics, and trajectory planning.
- Built and validated MATLAB and Simscape models and performed actuator/transmission analysis across payload cases from **0 to 1.0 kg**.
- Programmed TM collaborative robot tasks including pick-and-place, machine-vision detection, contact drawing, visor manipulation, hinge imaging, and stamping at multiple speeds and orientations.

GitHub: [industrial-robotics-visor-automation](#)

EduExo Brick Breaker – Transparent & Assist-as-Needed Control

MSc Academic Project

- Individually redeveloped an EduExo–Unity rehabilitation-style exergame with active transparent control, real-time tracking feedback, and configurable assist-as-needed intervention.
- Implemented **4 assistance levels** (Off, Low, Medium, High), automatic CSV performance logging, and timed **60-second** sessions.
- Conducted technical evaluation with **2 healthy users**, analysing tracking error, assistance activation, transparency, and independent movement across assistance settings.

[GitHub: eduexo-brick-breaker-rehabilitation](#)

Sign-Language Educational Robotic Hand

BSc Thesis

- Designed and fabricated a cable-driven compliant continuum wrist extending a commercial robotic hand for sign-language education.
- Developed SolidWorks CAD, Adams multibody simulations, additive-manufactured components, Raspberry Pi servo control, and a preliminary vision-based sign-selection proof of concept.
- Thesis received the **maximum grade and Excellence distinction**; work published with a permanent DOI.

[GitHub](#) • [DOI](#)

PHOENIX Bioinspired Robotic Fish

MSc Academic Project

- Designed, fabricated, and experimentally validated an Atlantic-mackerel-inspired robotic fish with a segmented tail, waterproof structure, embedded sensing, PID propulsion control, and IMU-based head stabilization.
- Contributed to tail CAD, 3D printing, silicone molding, shaft fabrication, mechanical assembly, electronics selection, sealing, and in-water testing.
- Demonstrated **0.33 m/s** swimming speed at 1.5 Hz (**0.73 body lengths/s**) and measured a **1.38 N** peak dynamic load during thrust testing.

[GitHub: bioinspired-robotic-fish](#)

WORK EXPERIENCE

CERN Summer Student – ATLAS RPC Prototype Testing

Jun 2023 – Aug 2023

CERN

Geneva, Switzerland

- Performed experimental characterization of the first RPC-BIL prototype for the ATLAS Muon Spectrometer upgrade in preparation for the Mechanics Final Design Review.
- Measured strip-line signal propagation velocity of **20.7 cm/ns** and established an approximate **2 cm spatial-resolution upper limit** using time-of-flight data.
- Studied detector hit-rate uniformity under different gas-volume pressure conditions and compared experimental distributions with simulation.

[GitHub](#) • [CERN Publication](#)

Technical Intern – Sensor Testing & Evaluation

Aug 2023 – Sep 2023

Thermorex CJSC

Yerevan, Armenia

- Performed hardware testing, sensor-output validation, and electrical checks on industrial temperature-control equipment.
- Documented test procedures, component faults, and operational measurements to support quality assurance.

Front-End Developer

May 2021 – Jan 2022

Jybird Communications

Yerevan, Armenia

- Developed responsive user-interface components using React, JavaScript, HTML, and CSS and integrated interfaces with backend REST APIs.

Communication & Support Assistant

Jun 2024 – Aug 2024

KASA Fondation Humanitaire Suisse

Yerevan, Armenia

- Coordinated digital communications, documentation, and logistical support for multilingual foundation initiatives.

EDUCATION

MSc in Robotics & Mechatronics Engineering

Sep 2024 – Present

Politecnico di Milano

Milan, Italy

- Visiting MSc thesis research at **University College London (UCL)**, London.
- Relevant coursework: Advanced Robotics, Multivariable Control, Real-Time Systems, Mechatronic Systems Design, Collaborative Robotics, Bioinspired Robotics.

BSc in Robotics & Mechatronics Engineering

Sep 2020 – Jun 2024

National Polytechnic University of Armenia

Yerevan, Armenia

- **GPA: 3.97 / 4.00** • **Red Diploma (Highest Honours)**
- **BSc Thesis: Maximum Grade** • **Excellence Distinction** — Sign-Language Educational Robotic Hand